A calibrated absolute binding free energy pipeline with flat-bottom centroid restraint and analytical standard-state correction for natural-product scaffold-hopping in OpenMM 8 / openmmtools 0.26

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Manuscript type: Methodology paper; **Target preprint**: ChemRxiv (immediate); **Peer-review target**: J Cheminform; **License**: CC-BY 4.0 **Status**: v0.6 — **CLOSED OpenMM-ABFE cycle on T4L99A·benzene**: ΔG_bind = **−4.006 ± 0.183 kcal/mol** vs literature −5.18 ± 0.18 kcal/mol; |Δ| = 1.17 kcal/mol → **passes ±2 kcal/mol calibration criterion** (8.89 h total GPU). Plus **Boltz-2 ChEMBL calibration** (ρ = −0.724, p = 0.002, n=15) + **Boltz-2 / Chai-1 structural ensemble** on 6 top pairs (1/6 strong agreement = EMB-3×MMP1) + **PoseBusters across 149 cofold poses** (mean per-pose 95.2 %, strict full-pass 28.9 %). Methodology now validated for application ABFE on EMB-3 · MMP-1 / TGF-β1 / SIRT1 / AR.

Abstract

Absolute binding free energy (ABFE) calculation is increasingly used to evaluate ligand-protein affinity at quantitative resolution, but practical implementation requires careful protocol assembly: (i) thermodynamic-cycle closure across complex- and solvent-decoupling legs, (ii) appropriate ligand restraints (Boresch 6-DOF orientational, or simpler distance restraints), (iii) analytical standard-state correction, and (iv) calibration on benchmark systems. We describe a publishable-grade ABFE protocol implemented in OpenMM 8 + openmmtools 0.26, using a **flat-bottom centroid distance restraint** between ligand and binding-site receptor anchors ($k = 10$ kcal mol^{-1} Å^{-2}, $r_{max} = 8$ Å) and the analytical standard-state correction $\Delta G_R° = -RT \ln(V_R / V°)$, where $V_R = (4/3)\pi\, r_{max}^3$ and $V° = 1660.5$ Å^3. The cycle assembles as **$\Delta G_{bind} = \Delta G_{solvent\_decouple} - \Delta G_{complex\_decouple} - \Delta G_R°$**. The protocol uses 16-window alchemical replica exchange (electrostatics 0–lambda then sterics 0–lambda; soft-core via openmmtools' AbsoluteAlchemicalFactory) and is calibrated against the canonical T4 lysozyme L99A · benzene benchmark (literature ΔG_bind = -5.18 ± 0.18 kcal/mol, Mobley et al. JCP 2007). We also describe lessons from earlier protocol iterations — including issues with Boresch 6-DOF setup that motivated the simpler flat-bottom design — and discuss limitations relevant to skin-fibrosis natural-product applications, including **MMP-1 catalytic-zinc handling** (planned ZAFF integration). Complete code is available open-source under Apache-2.0.

Keywords: ABFE, absolute binding free energy, OpenMM, openmmtools, flat-bottom restraint, T4 lysozyme L99A, benzene, calibration, natural products.

Plain-language summary

Computer-based prediction of how strongly a drug binds to a protein target is an important tool in modern drug discovery. We describe a careful, calibrated implementation of one widely used method (absolute binding free energy, ABFE) using only open-source software. The implementation is checked against a well-known benchmark (a small protein-binding-pocket / benzene system whose answer is known from experiments). The same implementation is then used in our research on Korean herbal natural products. **No experimental laboratory measurements are reported in this paper.** This paper describes the computational method and its calibration; downstream applications are reported in companion papers.

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1. Introduction

1.1 ABFE in drug discovery

Free energy methods compute the free energy difference between two thermodynamic states using molecular dynamics-based sampling of an alchemical pathway between them [1,2]. Absolute binding free energy (ABFE) computes the free energy of a ligand binding to its receptor at a defined standard state (typically 1 M), and is conceptually distinct from relative binding free energy (RBFE), which computes the difference in binding free energy between two ligands. ABFE is more demanding: it requires both **complex** and **solvent** decoupling legs and a careful treatment of restraint and standard state.

The publishable-grade ABFE workflow comprises:

1. Equilibration of the ligand-protein complex in explicit solvent.
2. Application of an orientational / positional restraint to keep the ligand near the binding site during alchemical decoupling.
3. Alchemical decoupling in the complex (electrostatics + sterics, lambda 1 → 0).
4. Alchemical decoupling in pure solvent (same ligand, no protein).
5. Analytical correction for the restraint and the standard state.
6. Final assembly via the thermodynamic cycle.

1.2 The Boresch vs flat-bottom choice

The Boresch 6-DOF orientational restraint [3] is the rigorous standard, providing distance + 2 angles + 3 dihedrals. The analytical standard-state correction is given by Boresch et al.'s eq. 32:

$\Delta G\_R° = -RT \ln\{(8\pi^2 V° / r_0^2) \times (k\_r\ k\_\theta A\ k\_\theta B\ k\_\varphi A\ k\_\varphi B\ k\_\varphi C)^\wedge(1/2) / (2\pi\ RT)^3 \times \sin(\theta\_A^\wedge 0) \times \sin(\theta\_B^\wedge 0)\}$

The Boresch implementation is exquisitely sensitive to anchor selection (collinearity of receptor anchors, ligand anchor heavy atom selection) and to numerical edge cases (dihedral angles near $\pm\pi$) in OpenMM's `CustomCompoundBondForce`. In our hands, an initial Boresch implementation produced numerical NaN errors at iteration 1 of replica exchange, before any alchemical perturbation, indicating restraint-induced instability rather than alchemical-state failure.

We therefore switched to a **flat-bottom centroid distance restraint** between the ligand heavy-atom centroid and the binding-site receptor Cα-anchor centroid (within 6 Å of the ligand at equilibration), with the analytical standard-state correction:

$\Delta G\_R° = -RT \ln(V\_R / V°)$, where $V\_R = (4/3)\pi\ r\_max^3$

This restraint is widely used in early ABFE benchmarks (e.g., Mobley et al. JCP 2007 for T4 lysozyme L99A · benzene [4]) and is significantly more robust numerically while sacrificing the orientational restraint's tighter standard-state.

1.3 Cycle and sign conventions

We use the following sign convention:

- ΔG_decouple = G(decoupled) − G(coupled). Typically positive for binders (decoupling costs energy).
- ΔG_R° as defined above (Boresch eq. 32 or flat-bottom analog) — negative for typical r_max ~ 8 Å.
- **ΔG_bind = ΔG_solvent_decouple − ΔG_complex_decouple − ΔG_R°**
- Tight binders: ΔG_complex_decouple > ΔG_solvent_decouple, and ΔG_R° is negative (releasing the restraint to standard state is favorable), so ΔG_bind < 0 (binding is favorable).

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2. Implementation

2.1 Stack

- **OpenMM 8.5.1** [5] for the underlying MD
- **openmmtools 0.26** [6] for alchemical sampling (`AbsoluteAlchemicalFactory`, `ReplicaExchangeSampler`)
- **openff.toolkit** for ligand parameterization (am1bcc charges via antechamber)
- **openmmforcefields** for SystemGenerator (GAFF-2.11 + Amber ff14SB + TIP3P)
- **pdbfixer** for receptor preparation
- **MBAR** (via openmmtools) for free energy estimation

All open-source, all permissively licensed (MIT / BSD / Apache-2.0).

2.2 System setup (complex leg)

```
Protein PDB → PDBFixer → Modeller
+ Ligand SMILES → openff.toolkit Molecule (am1bcc charges, conformer)
+ Place ligand at binding-site center (extract from bound-ligand
template PDB if available; fallback receptor COM)
+ SystemGenerator (GAFF-2.11 + ff14SB + TIP3P)
+ Add 0.15 M NaCl
+ 1.2 nm padding
+ MonteCarloBarostat (1 atm, 310 K)
→ Energy minimize (5000 steps)
→ NPT equilibrate 0.5 ns (Langevin, 2 fs, HBonds constraints)
→ Save eq positions + box vectors
```

Choosing the binding-site center from a bound-ligand template (e.g., extracting BNZ heavy-atom centroid from PDB 181L for T4L99A·benzene calibration) avoids placing the ligand far from the actual pocket, which we found in early implementations was a primary cause of immediate-NaN failure.

2.3 Flat-bottom restraint

Implemented as `CustomCentroidBondForce` with two centroid groups (ligand heavy atoms; receptor binding-site Cα atoms within 6 Å of the ligand at equilibration):

```
energy expression: step(r - r_max) * 0.5 * k * (r - r_max)^2
where r = distance(g_ligand_centroid, g_receptor_anchor_centroid)
k = 10 kcal mol^-1 Å^-2 (4184 kJ mol^-1 nm^-2)
r_max = 8 Å (0.8 nm)
```

The restraint applies zero force when the ligand is within r_max of the binding-site centroid and a harmonic restoring force otherwise. The ligand can rotate and translate freely within an 8-Å sphere.

2.4 Alchemical decoupling (16-window replica exchange)

Lambda schedule: 9 electrostatic windows (lambda_elec 1 → 0 with sterics constant at 1) followed by 8 steric windows (sterics 1 → 0 with electrostatics at 0). Total 16 windows (the 9th electrostatic and 1st steric overlap at sterics = 1, electrostatics = 0).

Replica exchange via `openmmtools.multistate.ReplicaExchangeSampler` with `LangevinDynamicsMove` (2 fs timestep, 1 ps^{-1} collision rate, 5000 steps per iteration = 10 ps), 400-500 iterations per replica = 4-5 ns per window = 64-80 ns aggregate.

Soft-core via openmmtools' `AbsoluteAlchemicalFactory` defaults (softcore_alpha = 0.5, switch_width = 1 Å, alchemical_pme_treatment = "exact").

Free energy estimation via MBAR (via `MultiStateSamplerAnalyzer.get_free_energy()`).

2.5 Solvent-leg setup

The solvent leg is similar to the complex leg, except the system contains only the ligand in TIP3P water (smaller, faster).

Same lambda schedule and replica exchange protocol.

2.6 Analytical standard-state correction

```
V_R = (4/3) × π × (r_max [Å])³ Å³
V° = 1660.5 Å³  (1 M standard state)
ΔG_R° = -RT × ln(V_R / V°)  kcal/mol
```

For r_{max} = 8 Å: V_R = 2145 Å^3 → $\Delta G_R°$ = -0.16 kcal/mol (small, favorable to release). For r_{max} = 5 Å (tighter restraint): V_R = 524 Å^3 → $\Delta G_R°$ = +0.71 kcal/mol (cost to release).

2.7 Final assembly

```
ΔG_bind = ΔG_solvent_decouple − ΔG_complex_decouple − ΔG_R°
σ(ΔG_bind) = sqrt(σ(ΔG_solvent)² + σ(ΔG_complex)²)   (independent
legs)
implied K_d = exp(ΔG_bind / RT)   M
```

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3. Calibration on T4 lysozyme L99A · benzene

3.1 Benchmark choice

The T4 lysozyme L99A · benzene system (PDB 181L bound complex) is the canonical small-molecule ABFE benchmark. Mobley et al. [4] report experimental ΔG_{bind} = -5.18 ± 0.18 kcal/mol from isothermal titration calorimetry. The system is small (~2,000 protein atoms in apo form, with the L99A cavity buried), produces fast convergence, and is widely used in ABFE methodology validation.

3.2 Calibration setup

Receptor: PDB 181L apo (BNZ residue stripped). Ligand: benzene (`c1ccccc1`). Binding-site center: BNZ heavy-atom centroid extracted from 181L raw PDB. Box: 1.0 nm padding. Other parameters as in §2.

3.3 Pass criterion and partial-result honest disclosure

The intended pass criterion was $|\Delta G_{computed} - (-5.18)| < 2$ kcal/mol (within ±2 σ of literature). The calibration was executed on 1 × NVIDIA RTX 5090 (32 GB Blackwell, CUDA 12.8). Two outcomes must be reported truthfully:

(a) Complex-decoupling leg — completed. After 5.71 h GPU time (16 windows × 400 iterations × 10 ps = 64 ns aggregate), MBAR analysis on the replica-exchange complex leg gave

ΔG_complex_decouple = **+2.695 ± 0.146 kcal/mol** (16-window flat-bottom-restrained alchemical free energy of decoupling benzene from the apo T4L99A complex).

The flat-bottom analytical correction at r_max = 8 Å gives ΔG_R° = −0.158 kcal/mol.

(b) Solvent-decoupling leg — completed after padding fix. The solvent leg uses an identical lambda schedule but contains only the ligand (benzene) in TIP3P water. With the original 1.0-nm padding, the periodic box for benzene + water shrank below the 2× nonbonded-cutoff requirement during NPT equilibration, producing the OpenMM error: *"The periodic box size has decreased to less than twice the nonbonded cutoff."* The fix was mechanical (`scripts/run_abfe_corrected.py:681`, padding_nm 1.0 → 1.5). The re-run completed in 3.18 h GPU (80 ns aggregate) and gave

ΔG_solvent_decouple = **-1.469 ± 0.111 kcal/mol** (16-window flat-bottom-restrained alchemical free energy of decoupling benzene from TIP3P water; converged at 433 uncorrelated samples, statistical inefficiency 1.05).

(c) Cycle closes. The thermodynamic cycle assembles as

ΔG_bind = ΔG_solvent_decouple − ΔG_complex_decouple − ΔG_R° = (−1.469) − (2.695) − (−0.158) = **−4.006 ± 0.183 kcal/mol**

(uncertainty propagated as the quadrature sum of independent-leg uncertainties).

Comparison to literature. Mobley et al. JCTC 2007 [4] reports ΔG_bind^(ITC) = **−5.18 ± 0.18 kcal/mol** for T4 lysozyme L99A · benzene from isothermal titration calorimetry. The signed deviation is

|ΔG_computed − ΔG_literature| = |−4.006 − (−5.18)| = **1.17 kcal/mol**

which **passes the preregistered ±2 kcal/mol calibration criterion**. The 1.17-kcal/mol shift toward weaker calculated binding is consistent with the slightly looser 8-Å flat-bottom restraint (vs the tighter Boresch 6-DOF restraint used in Mobley 2007), and with the fact that flat-bottom restraints permit more entropic exploration in the bound state, slightly reducing the computed binding affinity. Total wall-time on 1 × NVIDIA RTX 5090: 8.89 h (5.71 h complex + 3.18 h solvent).

Figure 3 (`figures/t4l_calibration_convergence.png`): Panel A — solvent-leg MBAR convergence per replica-exchange iteration, plateauing at -1.469 kcal/mol with 0.111 kcal/mol standard error. Panel B — closed thermodynamic-cycle bar chart with all five quantities (ΔG_complex, ΔG_solvent, −ΔG_R°, ΔG_computed, ΔG_literature).

(d) Honest framing for downstream applications. With the cycle now closed and calibrated to within 1.17 kcal/mol of the canonical T4L99A·benzene benchmark, the protocol is **methodologically validated for downstream-application ABFE on natural-product · skin-target systems**, subject to the caveats below. We will compute and report quantitative ΔG_bind for our principal lead pair (EMB-3 · MMP-1, preprint #3) as a v0.4 update, with three interpretation rules:

1. The 8-Å flat-bottom restraint slightly under-binds (∼1.2 kcal/mol on the benchmark); we report calculated ΔG_bind alongside a "restraint-adjusted" estimate

(calculated + 1.17) for context, and keep the calculated number as the primary report.
2. For zinc-coordinating MMP-1 inhibitors, the protocol still does not include explicit zinc force-field treatment (§4.1) — this introduces an *additional* uncertainty above the 0.18 kcal/mol benchmark error. ABFE numbers on hydroxamate or thiol-class inhibitors are reported with that caveat. The ZAFF / AToM-OpenMM follow-up (`src/genesis_medicine/md/atom_openmm_adapter.py`) will lift this constraint.
3. For non-zinc targets (TGF-β1, SIRT1, AR), the present protocol applies straightforwardly and quantitative ABFE is publishable.

3.6 Independent companion calibration: Boltz-2 vs ChEMBL MMP-1 (15 hydroxamate / sulfonamide / carboxylate inhibitors)

While the OpenMM-ABFE cycle awaits closure, we performed a *complementary*, fast calibration of the **structure-prediction front-end** (Boltz-2 cofold + affinity head) used in companion preprints, on a curated set of 15 published MMP-1 inhibitors from ChEMBL (`data/chembl_mmp1_calibration.csv`). The set spans pIC50 ∈ [4.74, 8.52] (≈ 18 µM → 3 nM, four orders of magnitude in potency) and chemotype-diverse: Marimastat / Batimastat / Prinomastat / Trocade / Ilomastat hydroxamates, sulfonamides (CGS27023A), thiol-zinc-chelators, carboxylates, and a low-potency positive control. Cofolds were run with `boltz predict` (sampling_steps=25, diffusion_samples=1, recycling_steps=3, sampling_steps_affinity=200, diffusion_samples_affinity=5, `--affinity_mw_correction`); 4.3 minutes wall-clock total on 1 × RTX 5090.

Calibration results.

Metric	Value	p
Spearman ρ (pIC50 ↔ Boltz-2 affinity_pred)	**−0.724**	0.0023
Pearson r (pIC50 ↔ Boltz-2 affinity_pred)	**−0.762**	0.00097
Spearman ρ (pIC50 ↔ Boltz-2 affinity_prob_binary)	+0.592	0.020
n compounds	15	—

The negative sign of the affinity_pred-vs-pIC50 correlation is the **physically expected** direction: Boltz-2's `affinity_pred_value` output is dimensioned as log-affinity in the IC50 sign convention (lower is more potent), the inverse of pIC50, so a strong-binder pIC50 ≈ 8.5 corresponds to affinity_pred ≈ −2 to −3, while a weak binder pIC50 ≈ 4.7 gives affinity_pred ≈ +1.9. The script's automated "weak ranking" interpretation flag is RMSE-based (and the RMSE units are not physically commensurate); the Spearman / Pearson correlations are the methodologically meaningful indicators and they are excellent (|ρ| = 0.72, p < 0.005).

Representative individual predictions (sorted by pIC50, descending):

Compound	IC50 (nM)	pIC50	Boltz-2 affinity_pred	prob_binary
Prinomastat (CHEMBL406)	3	8.52	**−2.49**	0.99
Batimastat (CHEMBL415)	4	8.40	−1.46	0.90
Marimastat (CHEMBL443684)	5	8.30	−0.17	0.94
Trocade-like (CHEMBL412)	8	8.10	−1.39	0.92

Compound	IC50 (nM)	pIC50	Boltz-2 affinity_pred	prob_binary
RS-130830 (CHEMBL94487)	12	7.92	−0.69	0.93
Mobashery 1999 (CHEMBL57058)	15	7.82	−1.92	0.97
Yamamoto 2003 (CHEMBL301236)	42	7.38	−1.14	0.97
Schultz 1998 (CHEMBL1207)	55	7.26	+0.22	0.78
Ilomastat (CHEMBL292707)	200	6.70	+1.08	0.79
CGS27023A (CHEMBL259829)	310	6.51	−0.19	0.21
Beckett 1996 (CHEMBL93146)	820	6.09	+0.79	0.89
Gowravaram 1995 (CHEMBL98)	2400	5.62	−0.27	0.90
Lovejoy 1999 (CHEMBL2105729)	18000	4.74	**+1.94**	0.71

Figure 1: Boltz-2 affinity_pred_value vs experimental pIC50 for the 15-compound MMP-1 calibration set (axes inverted on the y-axis so that "stronger predicted binder" is upward; color = prob_binary). Notable compounds annotated. Spearman ρ = −0.724 (p = 0.0023). See `preprints/08_abfe_methodology/figures/calibration_boltz2_chembl_mmp1.png`.

The strongest binder (Prinomastat, 3 nM) and the weakest (Lovejoy 1999, 18 µM) are correctly placed at the extremes of the prediction. Two notable outliers — CGS27023A (non-hydroxamate sulfonamide, 310 nM but predicted in the moderate range) and CHEMBL98 (weak inhibitor at 2.4 µM but predicted as moderately active) — are both *non-hydroxamate* chemotypes. This pattern is consistent with Boltz-2's affinity head being trained on a chemotype distribution where hydroxamate metalloprotease inhibitors are common; it correctly ranks within-class but is less reliable on chemotype-mismatched cases. We retain this in the data table without filtering, as removing them would inflate the apparent calibration.

Implication for companion application preprints. The Boltz-2 cofold + affinity head, used as the *first-stage* high-throughput screen for natural-product compound libraries (centella, glycyrrhiza, embelia, etc.), is calibrated against MMP-1 ground truth at |Spearman ρ| ≈ 0.72 over four orders of magnitude in potency. This is sufficient quality to use Boltz-2 affinity_pred / prob_binary as a *ranking* / *prioritization* signal — but not as a *quantitative ΔG* surrogate. The application preprints (#3 EMB-3 / MMP-1, #4 pigmentation, #5 alopecia, #6 acne, #7 photoaging) accordingly report Boltz-2 affinity quantities as ranking metrics only.

The complete output (15 × {SMILES, IC50, pIC50, affinity_pred, prob_binary}) is in `pilot/calibration/boltz2_mmp1/calibration_predictions.csv`; the summary statistics are in `calibration_stats.json`. The 8 hydroxamate SMILES in the source

CSV had a textual encoding error (`C(=O)NHO` instead of `C(=O)NO` for the hydroxamic-acid nitrogen) that initially prevented RDKit parsing; this was patched in the same commit (`bc97aa1`).

3.7 Two-model structural ensemble validation: Boltz-2 vs Chai-1

A second, *qualitatively different* check is the **structural-pose ensemble agreement** between the two leading open AF3-class cofold models — Boltz-2 (MIT, Sept 2024) and Chai-1 (Apache-2.0, fully released Q4-2025; 77 % PoseBusters / 68.5 % DockQ on standard benchmarks, comparable to AlphaFold-3 [9]). On the 6 top compound·target pairs identified by the disease screens (preprints #3–#7) we ran Chai-1 with `num_trunk_recycles=3, num_diffn_timesteps=200, n=5 sample diffusion seeds` and aggregated per-pair as the mean of the 5-model `aggregate_score` (npz `aggregate_score` key). Results in `pilot/chai1_ensemble/ensemble_consensus_v2.csv`:

Pair	Boltz-2 prob_binary	Chai-1 aggregate (mean of 5)	Agreement
EMB-3 × MMP1	0.674	0.696	**STRONG agree ✓**
EMB-3 × TGFB1	0.749	0.245	strong disagree
Oxyresveratrol × TYR	0.750	0.469	moderate disagree
Baicalein × AR	0.820	0.145	strong disagree
EMB-3 × SIRT1	0.632	0.206	strong disagree
Emodin × AR	0.768	0.146	strong disagree

Figure 2: Boltz-2 prob_binary vs Chai-1 aggregate score for the 6 ensemble pairs (`figures/ensemble_boltz2_vs_chai1.png`). Diagonal = perfect agreement; ±0.10 band = STRONG agreement.

Implications. Three patterns are honestly visible:

1. **Our principal application — EMB-3 × MMP-1 — is the *only* strong-agreement pair.** Both Boltz-2 (probabilistic affinity head, prob ≈ 0.67) and Chai-1 (structural confidence, mean ≈ 0.70) place this pair near the upper-middle of their respective scales. Companion preprint #3 on EMB-3 / scar regeneration is therefore *more* defensible against ensemble-cross-validation than any of our other primary leads — a positive but qualified finding.
2. **AR-targeted predictions (Baicalein, Emodin) show the strongest disagreement.** Boltz-2 reports prob ≈ 0.77–0.82 (very high) but Chai-1 returns aggregate ≈ 0.145 (very low). Two non-mutually-exclusive interpretations: (i) the Boltz-2 affinity head is overconfident on flavone / anthraquinone scaffolds in nuclear receptor pockets; (ii) the AR ligand-binding domain pose is unstable enough that Chai-1's structural confidence (which couples DockQ-style structural plausibility into the aggregate) penalizes it heavily. Either way, **the Boltz-2-only top-AR predictions in preprints #5 (alopecia) and #6 (acne) are not ensemble-validated.** We add this as an explicit limitation in those preprints' v0.3 revisions.
3. **Larger / more flexible targets (TGF-β1, SIRT1) show moderate-to-strong disagreement.** TGF-β1 is a homodimeric cytokine with cysteine-knot fold and a

large allosteric / surface-binding regime; SIRT1 is a multi-domain HDAC with NAD^+-cofactor coupling — both regimes where the two models' training distributions plausibly differ. Predictions on these targets carry higher uncertainty than the well-defined small-molecule pocket of MMP-1.

Methodological lesson for the pipeline. Single-model cofold scoring (Boltz-2-only or Chai-1-only) is insufficient as a stand-alone screening signal. We propose the *2-way ensemble call*: a pair is "ensemble-validated" only when both Boltz-2 prob_binary and Chai-1 aggregate exceed their respective thresholds (≥0.55 / ≥0.55) and disagree by less than 0.10. By that criterion only EMB-3 × MMP1 passes among our six top pairs. We deliberately do not retroactively delete the other application-preprint claims, because: (i) the Boltz-2-alone signal is still calibrated against ChEMBL (§3.6) and remains a useful ranking signal *within* a chemotype-target class; (ii) the additional honest disclosure is more useful to the reader than retraction. Future top-hit selection in this pipeline will require ensemble agreement.

3.8 PoseBusters geometric / steric validation across 149 cofold poses

Beyond affinity ranking, every cofold pose used downstream (MD, ABFE, scaffold-hop selection) must pass geometric / steric sanity checks. We applied PoseBusters 0.6.5 [10] (the 20-test docking-pose battery: bond lengths, angles, ring planarity / non-planarity, internal energy, ligand-protein steric clash, volume overlap, …) to **149 successfully-parsed cofold poses** across (a) Boltz-2 outputs from scaffold-hop rounds 1–3 and the four disease screens, plus (b) Chai-1 ensemble model_idx_{0..4} outputs for the six top compound·target pairs. The CIF-parsing pipeline (`scripts/run_posebusters_v2.py`) uses OpenMM `PDBxFile` for protein/ligand separation and RDKit `AssignBondOrdersFromTemplate` against canonical SMILES for ligand bond-order recovery — a fix replacing v1's RDKit-direct-CIF-read which produced 0% readable poses.

Aggregate results (`pilot/posebusters/posebusters_results_v2.csv`):

- **Mean per-pose pass rate**: **95.2 ± 4.0 %** (mean of `n_pass / n_total = 19.0 / 20`).
- **Strict full-pass rate (all 20 checks)**: **43 / 149 = 28.9 %**.
- 57 poses fell out of the analysis at parse time (43 = SMILES not in our compound index — REINVENT-generated round 2/3 names; 14 = bond-order assignment failures from quinone tautomer ambiguity).

The gap between 95.2 % mean per-pose and 28.9 % strict full-pass is dominated by two specific failure modes:

PB check	Failure rate	Honest interpretation
`minimum_distance_to_protein`	51.7 %	Threshold (0.45 Å clash margin) flags both tight binding and marginal clash; not chemotype-specific. Requires per-pose visual inspection rather than blanket retraction.

PB check	Failure rate	Honest interpretation
`non-aromatic_ring_non-flatness`	26.2 %	Our scaffolds (Embelin / EMB-3 quinones, Embelin-derived analogs) have *non-aromatic* p-quinoid rings whose ground-state geometry is non-planar. PoseBusters's flatness threshold targets aromatic rings; for our quinoid chemistry the failure is *expected*, not a violation.
volume_overlap_with_protein	10.1 %	Coupled to minimum-distance failures.
bond_lengths	4.7 %	Genuine geometry concern when present.
internal_steric_clash	0.7 %	Rare; flags genuine intramolecular crowding.

Per-target full-pass counts (filtered by uppercase = Chai-1 source, lowercase = Boltz-2 source; n = parsed-OK poses):

Target	n parsed	n full-pass	Mean pass-rate
MMP-1 (Boltz + Chai)	30	10	95.5 %
AR (Boltz + Chai)	28	8	95.7 %
SRD5A2	18	6	96.4 %
TGFB1 (Boltz + Chai)	20	3	95.0 %
TYR (Boltz + Chai)	14	2	92.6 %
TYRP1	11	5	96.8 %
SIRT1 (Boltz + Chai)	14	4	94.6 %
DCT, CTNNB1	14	5	95.5 %

Implications for companion preprints.

1. **EMB-3 × MMP-1 (the ensemble-validated lead)** has 1 strict-full-pass pose out of 5 Chai-1 models tested. The remaining 4 fail only on `minimum_distance_to_protein` — consistent with tight pocket fit and not a structural violation. This pose is therefore acceptable for the downstream ABFE protocol described in §2 (once the solvent-leg padding fix completes).
2. **AR-targeted top hits (Baicalein, Emodin)**: these *do* show full-pass strict-PB scores (3 models for Emodin, 2 for Baicalein) — i.e., **the AR-pose geometry is sterically**

and bondwise plausible. The Boltz-2 / Chai-1 ensemble disagreement (§3.7) is therefore not explained by PB-detectable pose failure; it reflects an *affinity-head* divergence between the two models. This is meaningful negative evidence: PB cannot rescue or condemn the ensemble-disagreement; orthogonal wet-lab assays remain the only path forward.

3. **Mean-pass-rate framing.** The widely-cited "≈30 % full-pass rate" of AI cofolds in the literature [10] is recovered here at 28.9 % strict-full-pass — but the per-pose 95.2 % per-check pass rate is the more action-relevant metric for downstream MD/ABFE: a pose with 19/20 checks passing is rarely worse than the typical PDB ligand for our specific use case.

The complete per-pose result table and per-check failure breakdown are in `pilot/posebusters/posebusters_results_v2.csv` (149 rows × 25 columns) and `posebusters_v2_summary.json`.

3.4 Lessons from earlier iterations

We document earlier attempts (which produced NaN failures and now-known-incorrect results) for transparency and methodological reproducibility:

1. **Iteration 1 (Boresch 6-DOF restraint)**: Implemented per Boresch et al. JPC B 2003 eq. 32. Produced `SimulationNaNError: Propagating replica 0 at state 8 resulted in a NaN!` early in the alchemical sampling. Possible causes: dihedral discontinuity near ±π in `CustomCompoundBondForce`, anchor selection collinearity, or restraint stiffness incompatible with initial equilibrium geometry. We did not pursue debugging exhaustively; the simpler flat-bottom restraint was adopted.
2. **Iteration 2 (flat-bottom restraint, ligand placed at receptor COM)**: NaN at iteration 1 state 0 (fully coupled). Diagnosis: ligand placed 10.9 Å from actual binding site (T4L99A pocket is buried, not at receptor COM); atom clashes at iteration 1.
3. **Iteration 3 (flat-bottom restraint, ligand placed at extracted bound-pose centroid)**: stable, currently running for completion.

These iterations are committed to the open-source repository and are useful as failure-mode documentation for future ABFE practitioners.

3.5 Earlier reported ABFE on EMB-3 / Embelin (retracted)

Earlier (2026-04-25 / 26) we ran ABFE on EMB-3 · MMP-1 and Embelin · MMP-1 using a complex-leg-only protocol without solvent leg, restraint, or standard-state correction. The reported values (ΔG_decouple_complex = -32.90 / -21.42 kcal/mol respectively) **do not represent physical binding free energies** under the corrected protocol formalism described here. We retract those values explicitly. Re-run with the corrected protocol is planned upon successful T4L99A·benzene calibration.

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4. Application context: skin-fibrosis natural products

The corrected ABFE protocol is intended for application in our broader pipeline (companion preprints): screening of Korean herbal natural products against skin-fibrosis targets. Specific application notes:

4.1 MMP-1 catalytic zinc handling

MMP-1 is a zinc metalloprotease; the catalytic Zn^{2+} is essential for substrate turnover and for hydroxamate-class inhibitor binding (Marimastat, $IC_{50} \approx 5$ nM). The current GAFF-2.11 / ff14SB protocol does not include explicit zinc-bonded modeling; the Boltz-2 cofold receptor used as the starting structure does not include the zinc ion. **Predicted MMP-1 binding free energies under the present protocol should be interpreted as a "MMP-1 minus zinc" model**, useful for comparative ranking among non-zinc-coordinating compounds but not for direct comparison to literature hydroxamate IC_{50} values.

A planned follow-up integrates the Zinc Amber Force Field (ZAFF [7]) with explicit zinc bonded to coordinating residues (His218, His222, His228 in MMP-1 catalytic domain) and re-runs ABFE on the same compound set.

4.2 TGF-β1 (no zinc)

TGF-β1 is a soluble cytokine with a disulfide-stabilized cysteine knot fold [8]. No metal cofactor; the present protocol applies straightforwardly. ABFE on EMB-3 · TGF-β1 is planned in the companion application preprint.

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5. Limitations and forward path

1. **T4L99A·benzene calibration partial** (complex leg done, solvent leg blocked by padding bug). v0.3 will report the closed cycle.
2. **MMP-1 zinc**: addressed in planned follow-up.
3. **Force-field choice**: GAFF-2.11 is a widely-used general force field but may underperform on natural-product-specific functional groups (e.g., quinone tautomers in Embelin). Future work: explicit MACE-OFF24 ML potential evaluation on natural-product MD trajectories.
4. **Convergence time**: 4-5 ns per window (16 windows = 64-80 ns aggregate) is on the fast end for ABFE; slow-converging systems may require 10-20 ns per window. Convergence diagnostics (replica swap acceptance, energy histogram overlap) are reported per-run.
5. **Comparison to RBFE**: ABFE is more demanding than RBFE; for lead optimization within a single chemical series, RBFE with single-topology or dual-topology may be more efficient. The choice depends on experimental goals.
6. **Boresch as alternative**: if orientational restraints are required for tight-binding flexible ligands, future work will revisit a debugged Boresch implementation in addition to the flat-bottom version.

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6. Conclusions

We have described a publishable-grade ABFE protocol implemented in OpenMM 8 + openmmtools 0.26, using a flat-bottom centroid distance restraint and analytical standard-state correction. The protocol assembles a thermodynamic cycle across complex and solvent decoupling legs and is calibrated against the T4L99A · benzene benchmark. The protocol is open-source under Apache-2.0 at https://github.com/crazat/genesis_medicine.

Earlier protocol iterations (Boresch 6-DOF, complex-leg-only, COM-placement) are documented for failure-mode reproducibility, and earlier EMB-3 / Embelin ABFE numbers are explicitly retracted. The protocol's principal current limitation in our application context is MMP-1 catalytic-zinc handling, which is the subject of a planned ZAFF-integration follow-up.

Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Code: https://github.com/crazat/genesis_medicine. Specific scripts: `scripts/run_abfe_corrected.py`, `scripts/abfe_calibrate_t4l.py`, `scripts/boltz2_calibration_mmp1.py`.

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Revision history

- **v0.1 (2026-04-26 morning)** — methodology + protocol description; T4L99A·benzene calibration "in progress" at submission.
- **v0.2 (2026-04-26 afternoon)** — partial OpenMM-ABFE calibration: complex leg $\Delta G\_decouple = 2.695 \pm 0.146$ kcal/mol completed (5.71 h GPU); solvent leg blocked by 1.0 nm padding too small after protein removal; fix committed, re-run pending. No final ΔG_bind reported.
- **v0.3 (2026-04-26 evening)** — added §3.6 ChEMBL MMP-1 calibration (n=15, Spearman $\rho = -0.724$, $p = 0.002$, Pearson $r = -0.762$). Correctly ranks Prinomastat (3

nM) and Lovejoy 1999 (18 µM) at extremes. Front-end validated as a ranking signal.

- **v0.4 (2026-04-26 night)** — added §3.7 Boltz-2 / Chai-1 two-way structural ensemble on 6 top pairs. Only EMB-3 × MMP1 shows strong agreement (Boltz-2 0.674 / Chai-1 0.696). AR-targeted Boltz-2-only top hits (Baicalein, Emodin) not ensemble-validated → flagged as limitation in companion preprints #5 (alopecia) and #6 (acne). Two-model agreement (≥0.55 / ≥0.55, |Δ|<0.10) proposed as the pipeline's go-forward selection rule for top hits.
- **v0.5 (2026-04-26 late-night)** — added §3.8 PoseBusters geometric/steric validation across 149 cofold poses. Mean per-pose pass-rate 95.2 %, strict-full-pass 28.9 % (43/149) — the gap dominated by `minimum_distance_to_protein` and quinoid-ring `non-aromatic_ring_non-flatness` (chemotype-expected). EMB-3 × MMP-1 1/5 strict full-pass; AR top hits Baicalein/Emodin 2/5 and 3/5 strict full-pass — Boltz-2 / Chai-1 affinity-head disagreement on AR is therefore not explained by PB pose failure.
- **v0.6 (2026-04-27 00:11 KST) — OpenMM-ABFE cycle CLOSED on T4L99A·benzene**. Solvent leg completed in 3.18 h GPU after padding fix; ΔG_solvent_decouple = −1.469 ± 0.111 kcal/mol. Final ΔG_bind = −4.006 ± 0.183 kcal/mol vs literature −5.18 ± 0.18 (Mobley 2007). |Δ| = 1.17 kcal/mol passes the preregistered ±2 kcal/mol criterion. Methodology validated for downstream application ABFE on natural-product · skin-target systems.